

**CENG381 Stochastic Processes
Midterm Exam**

Duration: 90 minutes

Q1 (35 points). I have 4 umbrellas, some at home, some in the office. I keep moving between home and office. I take an umbrella with me only if it rains. If it does not rain, I leave the umbrella behind (at home or in the office). It may happen that all umbrellas are in one place, I am at the other, it starts raining and must leave, so I get wet.

1. If the probability of rain is p , what is the probability that I get wet?
2. Current estimates show that $p = 0.6$ in Edinburgh. How many umbrellas should I have so that, if I follow the strategy above, the probability I get wet is less than 0.1 ?

Q2 (35 points). A machine is subject to failures of types $i = 1, 2, 3$ at rates λ_i and a failure of type i takes an exponential amount of time with rate μ_i to repair. Formulate a Markov chain model with state space $\{0, 1, 2, 3\}$ and find its stationary distribution. Solve the problem in the concrete case $\lambda_1 = 1/24$, $\lambda_2 = 1/30$, $\lambda_3 = 1/84$, $\mu_1 = 1/3$, $\mu_2 = 1/5$, and $\mu_3 = 1/7$.

Q3 (30 points). Five white balls and five black balls are distributed in two urns in such a way that each urn contains five balls. At each step we draw one ball from each urn and exchange them. Let X_n be the number of white balls in the left urn at time n .

Compute the transition probability for X_n either analytically or using computer-intensive statistical methods (Develop a pseudocode algorithm that simulates this stochastic process over N steps to estimate the transition probabilities empirically.).